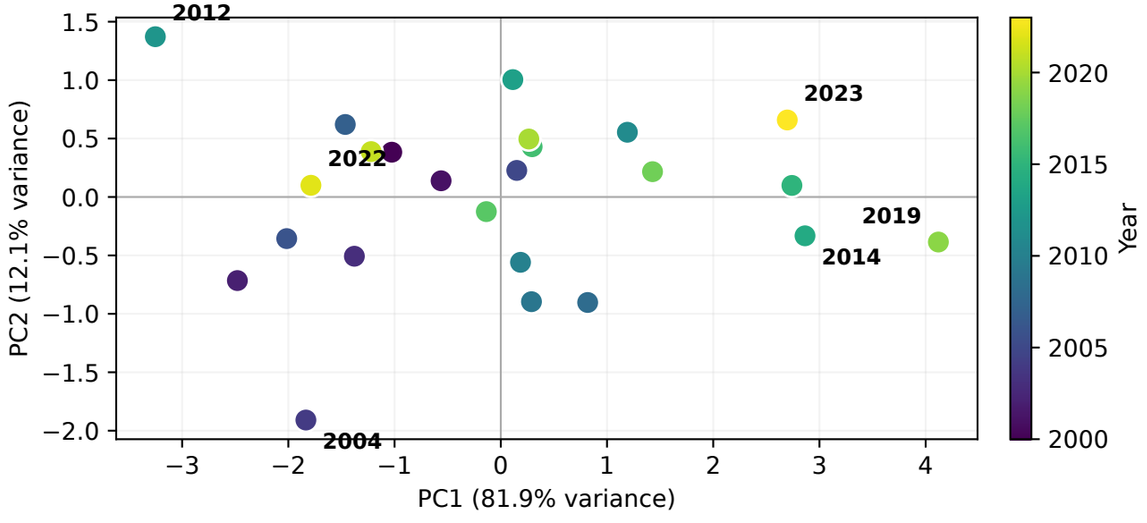
MULTIVARIATE ENVIRONMENTAL VARIABILITY

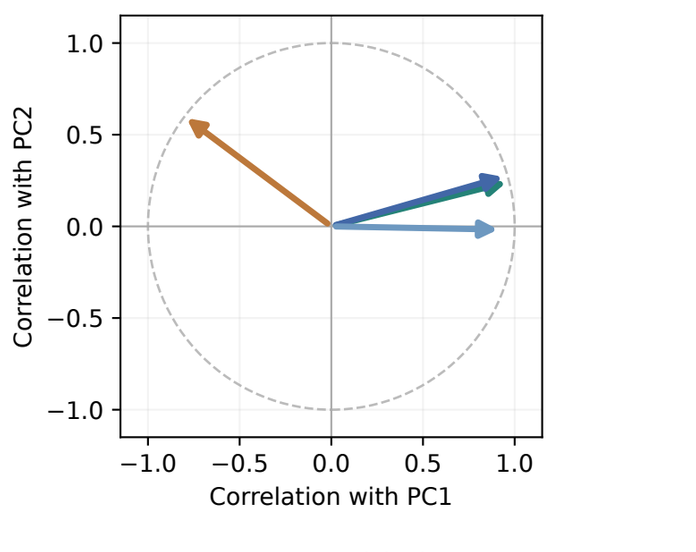
Environmental Data Cube for Oglala Lakota

Ate Makoce | May–September 2000–2023 | 24 annual observations | 4 environmental indicators

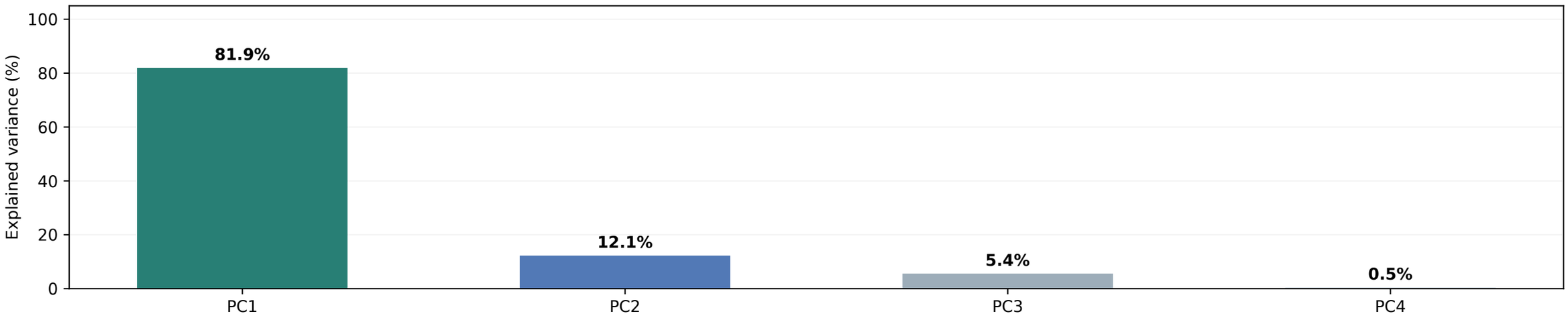
A | Annual environmental PCA scores



B | Environmental variable loadings



C | Variance explained by each component



RESULTS INTERPRETATION

The first two principal components explain 94.06% of total standardized variance (PC1: 81.95%; PC2: 12.11%). PC1 is positively associated with EVI, NDMI, and precipitation, and negatively associated with maximum temperature. These results describe a dominant vegetation–climate gradient across the observed growing seasons.

Interpretive limitation: PCA identifies associations rather than causes. The analysis does not establish overgrazing, land degradation, or management effects. Vegetation and regional climate measurements may represent different spatial footprints.

Source: Notebook 09, SA 15C.1 | EVI, NDMI, precipitation and maximum temperature, 2000–2023.